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Welcome to our **series on renewable resources**

The topic today is hydropower. As you most probably know, 'hydro' means water-so we are talking about using water to

generate electricity

. Of course, there are many ways to generate electricity but hydropower is

important to the community , firstly and obviously, because **it's renewable**

. The earth's hydrologic cycle of constant evaporation and transpiration

provides a continual supply of water from rainfall and snowmelt.

The second point to consider is its efficiency

Hydro power plants are able to convert approximately 90% of the energy from the falling water into electric energy whereas many fossil-fuelled plants

lose more than half of the energy content of

their fuel by way of waste heat and gases. For this reason they are very efficient.

Hydropower **is also clean** It doesn't emit **harmful harmful gases**

that contribute to air pollution, acid rain, and global warming. No trucks, trains or pipelines are needed to bring fuel to the site and

there's no noise pollution either

. Furthermore, hydropower plant machinery is fairly simple and **runs at slow speeds**

, which makes it reliable and durable. And hydropower units are flexible also—they have the ability to

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start quickly and adjust rapidly

to changes in demand for electricity, thus enabling them to

meet peak loads

. But this also allows them to serve as reserve capacity and

bring more stability to

the power system overall.

The dams that provide hydroelectric power also have other uses such as navigation, flood damage reduction, water supply, recreation, irrigation and low flow augmentation but

it's not the purpose of this talk to go into those details.

How do the hydropower plants work? Well,

a dam is built across a river

which captures water to form a reservoir and

raises the water level

to create 'head'. Think of 'head' as the vertical distance that the water falls as it passes through the dam, in other words, the difference in water level between the reservoir

behind the dam and the river below

. Water from the reservoir flows through an intake gate into a penstock—

this is kind of a narrow channel which leads to the turbine below. The force of the water

causes the turbine to rotate rapidly

which, in turn, drives the generator to spin and produce electricity. The electricity is carried the long distances

from the powerhouse to

subst



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of cities

via power lines

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Can you build a hydropower unit

on any river

? Well, no, just having water in a river isn't enough. A good dam site must

have enough stream flow

as well as enough head.

A fast flowing river

on the plains is probably not suitable because a dam couldn't be built high enough to provide the head needed

for efficient production of electricity

. On the other hand, dams in arid high country

may have plenty of

head but insufficient stream flow. The perfect spot for a hydropower plant is where the right combination of stream flow and head exists.

What about the environment

? Surely the construction of large dams

has an environmental impact

? Well, yes it does. Certainly dams and reservoirs

are built to improve the lives of people living in towns

, farming communities and cities. But there must be a balance between development and preserving the natural environment.

Needless to say

, the natural river environment is changed which leads to changes in river ecology and

get so low in summer that aquatic habitat

. Sometimes, for example, dissolved oxygen levels below dams

get so low in summer that

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there is a negative impact on aquatic life. These levels can be improved, however, by using special aerating turbines and /or injecting oxygen directly into the stream flow.

In order to protect and improve the habitat

for endangered and other species of birds, fish and water life, there needs to be a thorough review of operating plans to see if

a better balance can be achieved .

Hydropower plant design and operation

must not only meet the needs of consumers

for electricity but

work hand-in-hand with

agencies whose concern is for the fish and wildlife, water quality and water supply.